MBAI 5110: Big Data Systems Design

A Lambda Architecture Implementation for Predicting Bike Theft: A Case Study on Toronto Bicycle Theft

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Project Summary

This project showcases the design and implementation of a Lambda Architecture-based big data pipeline aimed at analyzing and predicting bicycle theft risk in Toronto. By integrating batch processing for historical analysis and model training with real-time streaming for live prediction, the system aims to provide actionable insights. A Random Forest classifier is utilized to categorize theft risk levels (Low, Medium, High), with careful attention given to preventing data leakage. This solution illustrates how scalable data systems can deliver valuable insights for both city stakeholders and individual users.

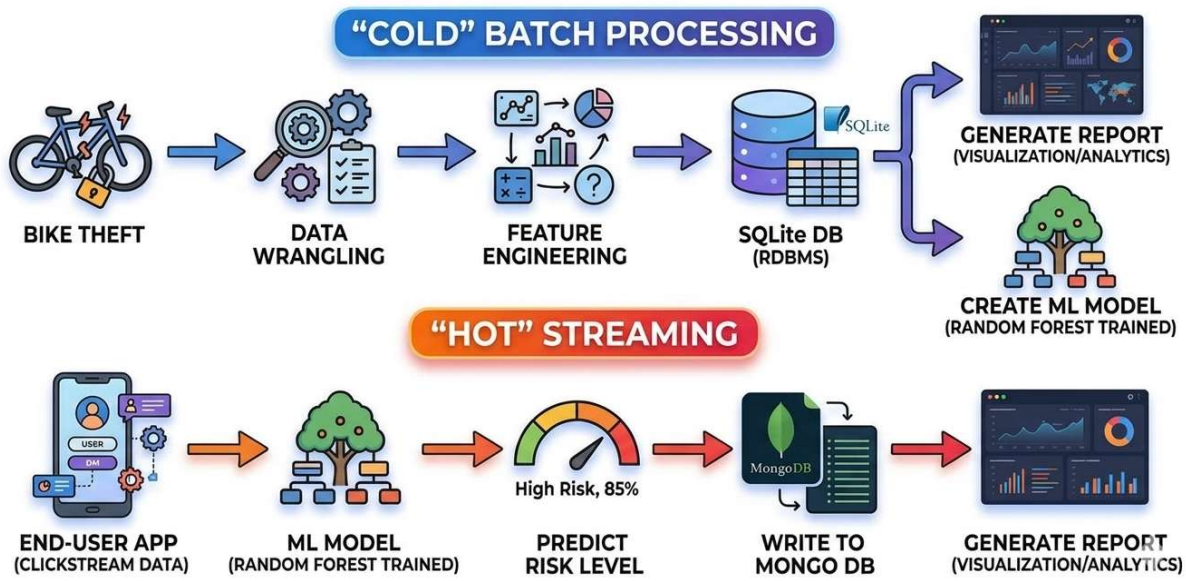
Dataset

The analysis utilizes the Bicycle Thefts dataset from the City of Toronto Open Data Portal (1975–2025).

- **Data Source**: City of Toronto Open Data Portal
- **Raw Data Size**: (39,849, 35)
- **Post-Data Wrangling Size**: (31,559, 33)

Data Pipeline Architecture

The architecture diagram below illustrates the Lambda Architecture designed to analyze bike theft patterns and provide real-time risk assessments. This structure effectively bridges the gap between historical "Cold" data and live "Hot" data, encompassing two phases: **Phase 1** (Batch Processing – "Cold") and **Phase 2** (Streaming - "Hot").



Phase 1: The "Cold" Path (Batch Processing)

This layer processes the "Master Dataset" to create highly accurate models and produce long-term reports. It focuses on handling large data volumes where processing time is less critical than precision.

1. **Ingestion**: The raw Bike Theft dataset is collected.

2. **Preparation**: Data preparation consists of two main parts:

    - **Data Cleanup**: This phase involves removing duplicates and addressing missing values through methods such as imputation and row deletion.

- **Feature Engineering**: Key variables are created (e.g., Occurrence Period, Risk Levels). For further details, refer to the Feature Engineering section of this report.

- **Storage**: The cleaned data is exported into a structured CSV file (Cleaned_Bikes.csv).

3. **RDBMS Management**: The processed data is imported into SQLite (RDBMS) for exploratory analysis, chosen for its effectiveness in structured, relational queries and generating fixed reports on historical data.

4. **Exploratory Analysis**: This phase involves a detailed inspection of the cold data:

    - **Identifying the Problem**: The 5 Ws framework (Who, What, When, Where, Why) provides a conceptual lens for understanding the dataset, making analysis more interpretable and actionable.

    - **Visualizing the Problem**: Generate reports to illustrate historical trends (DataVisualization_RDMS.ipynb).

5. **Intelligence**: Based on identified problems, a Random Forest model is trained to learn indicators predicting the Risk Level of bike theft. A Random Forest model is particularly effective as the classification problem demands precise risk categorization.

The Bridge: Random Forest Model

The Random Forest Model acts as the core of the system. Once trained in the Cold Path, it is deployed to the Hot Path for live decision-making.

Phase 2: The "Hot" Path (Real-Time Streaming)

This layer manages live data from the End-user App (Clickstream), optimized for low latency to provide instant risk assessments.

1. **Live Input**: The End-user App captures data such as bike colour, cost, make and model.

2. **Transmission**: Data is transmitted into the Hot Streaming pipeline.

3. **Real-time Prediction**: Live data is fed directly into the pre-trained machine learning model for instantaneous risk level predictions (e.g., “High Risk”).
4. **Integration and Serving**: The final stage converts raw predictions into actionable insights:
    - **Storage**: Results (Risk Level Determined) are stored in MongoDB, a NoSQL database adept at managing high-speed, unstructured clickstream data without slowing app performance.
    - **Output**: The system generates reports and visualizations, enabling stakeholders and users to access analytics on theft trends and current safety levels.

Key Insights

The key insights derived from the Historical Data Analysis (DataVisualization_RDMS.ipynb) reveal several noteworthy patterns:

1. **Temporal Insights**:
    - **Nighttime is Peak Time**: With 8,613 reported thefts, the night period slightly outnumbers morning and afternoon incidents. This indicates that thieves prefer the cover of darkness and reduced foot traffic for their activities. *Illustration: Chart 1: Bike Thefts by Time of Day and Chart 2: Bike Thefts by Hour of Day*

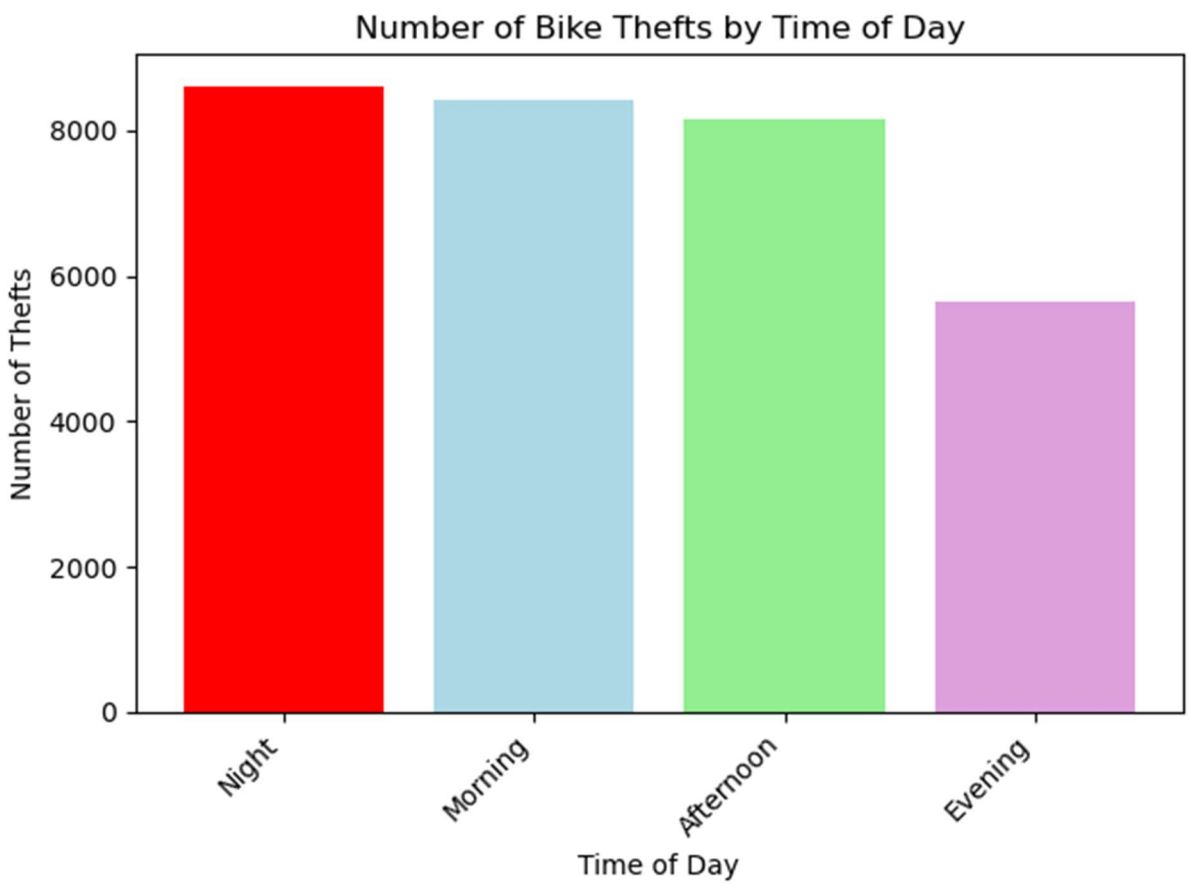


Figure 1- Bike Theft by Time of Day

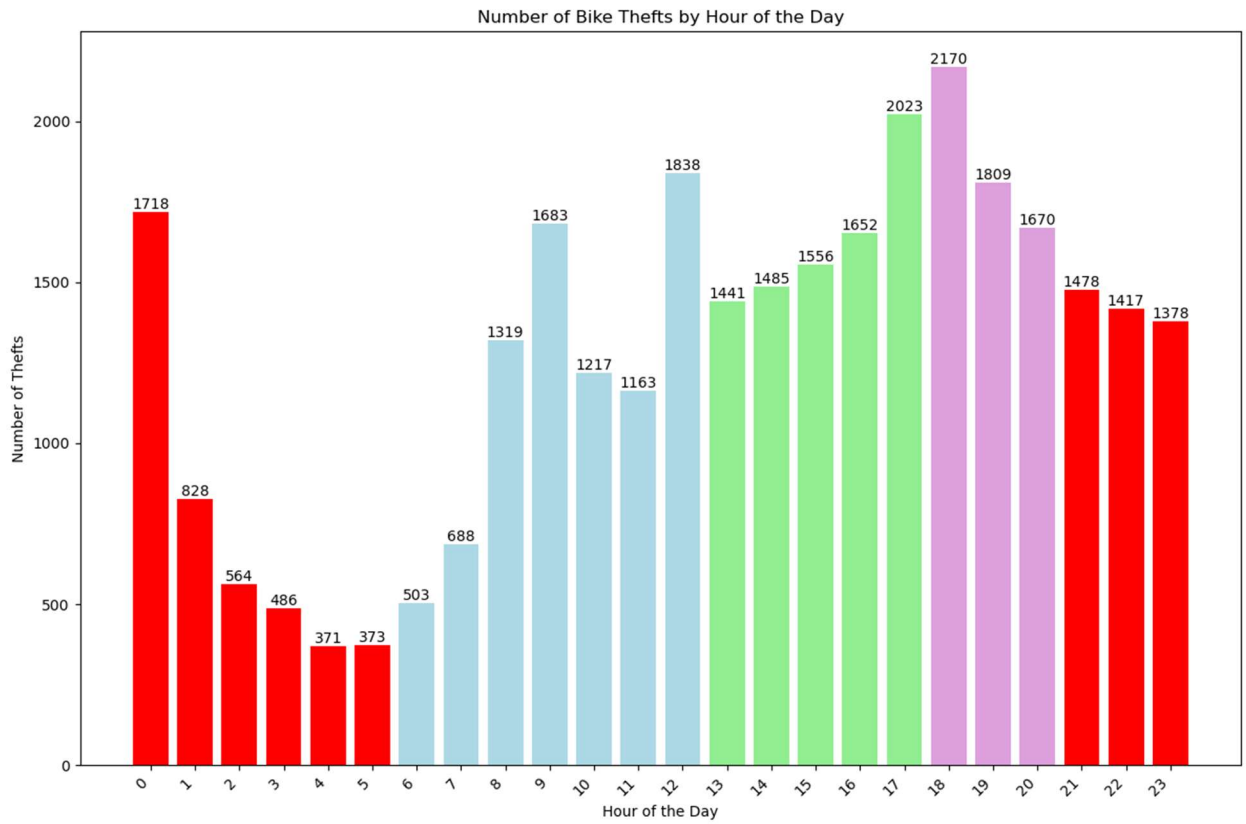


Figure 2- Bike Thefts by Hour of Day

2. **Spatial Insights**:
    - In the night category, thefts from Apartments/Condos (2,479) and Single Homes (1,865) dominant, highlighting the risks associated with private structures like sheds and garages (995), which are more frequently targeted

than bars or restaurants (245). *Illustration: Chart 3*- Bike Thefts in the Night by Location Type

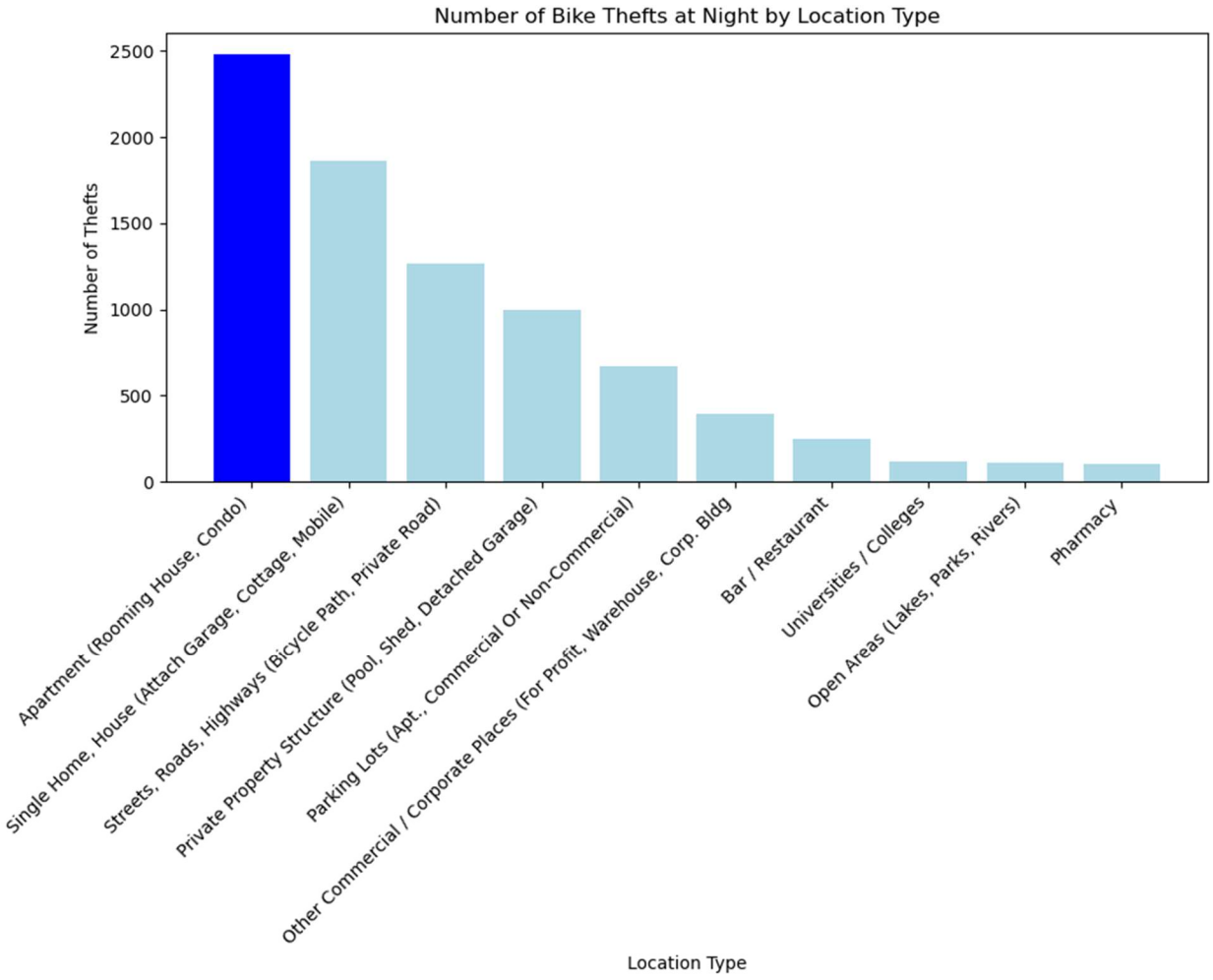


Figure 3- Bike Thefts in the Night by Location Type

3. **Geographic Hotspots**:
    - The Downtown Core shows a clear cluster of thefts, particularly in the Waterfront Communities (e.g., St. Lawrence, Harbourfront), likely due to dense modern condo developments with expansive underground parking.

Illustration: Chart 4- Concentration of Bike Thefts at Night in Apartment Complexes by Neighbourhood

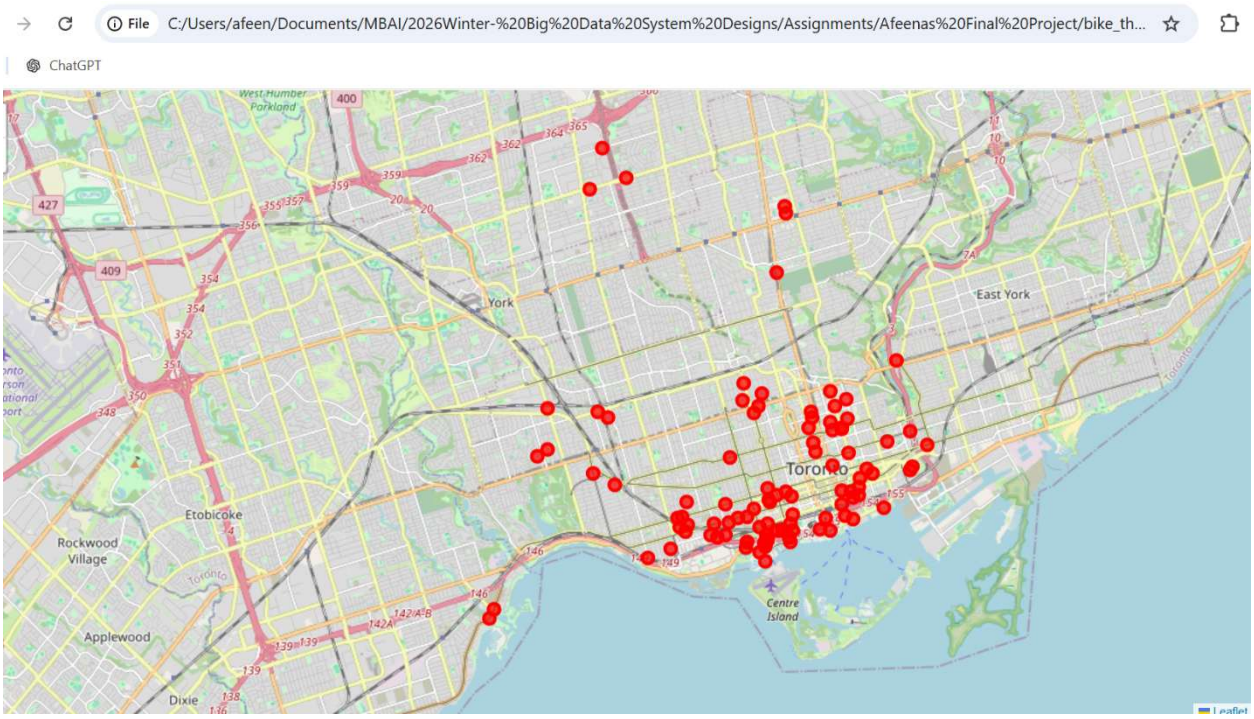


Figure 4- Concentration of Bike Thefts at Night in Apartment Complexes by Neighbourhood

Location	Bike Thefts
St Lawrence-East Bayfront-The Islands (166) Waterfront Communities-The Island (77)	147
Wellington Place (164) Waterfront Communities-The Island (77)	140
Harbourfront-CityPlace (165) Waterfront Communities-The Island (77)	135
Annex (95) Annex (95)	130
Fort York-Liberty Village (163) Niagara (82)	121

Table 1: Top 5 neighbourhoods by apartment-night theft count

Business Problem

The data indicates that despite residents investing in “secure” housing (condos and apartments), these locations account for the highest theft losses, leading to several secondary challenges:

- **Property Management Liability**: Heightened pressure on building managers to enhance surveillance and access controls.
- **Urban Mobility Barriers**: High theft rates in dense areas discourage residents from considering bicycles as viable transportation, contradicting city-wide “Green” initiatives.
- **Resource Misallocation**: Police efforts often focus on public areas, while data suggests that the frontline of bike theft has shifted to private spaces like parking garages.

Proposed Solutions

Based on insights identified above, several strategies can help curb bike thefts, categorized as **City Solutions** and **Individual Solutions.**

City Solutions

Given that the highest theft volumes occur at night, particularly around 6 PM (2,170 thefts), police should prioritize patrolling apartment complexes during the hours of 6 PM and midnight.

Individual Solutions

Residents currently lack a tool to accurately gauge risk when deciding where or how to store their bicycles. While the machine learning model was trained on over 30 historical features, the End-User Application focuses on "Live Risk Assessment." I simplified the user interface by:

1. **Removing Post-Facto Features**: Eliminating columns that are only relevant after a theft occurs (e.g., PRIMARY_OFFENCE, DIVISION).
2. **Automating Temporal Features**: Using the system clock to populate time-related fields.
3. **Focusing on User Inputs**: Users provide only essential details (make, model, cost, colour) and their current location, allowing the model to compare against historical patterns and output risk levels.

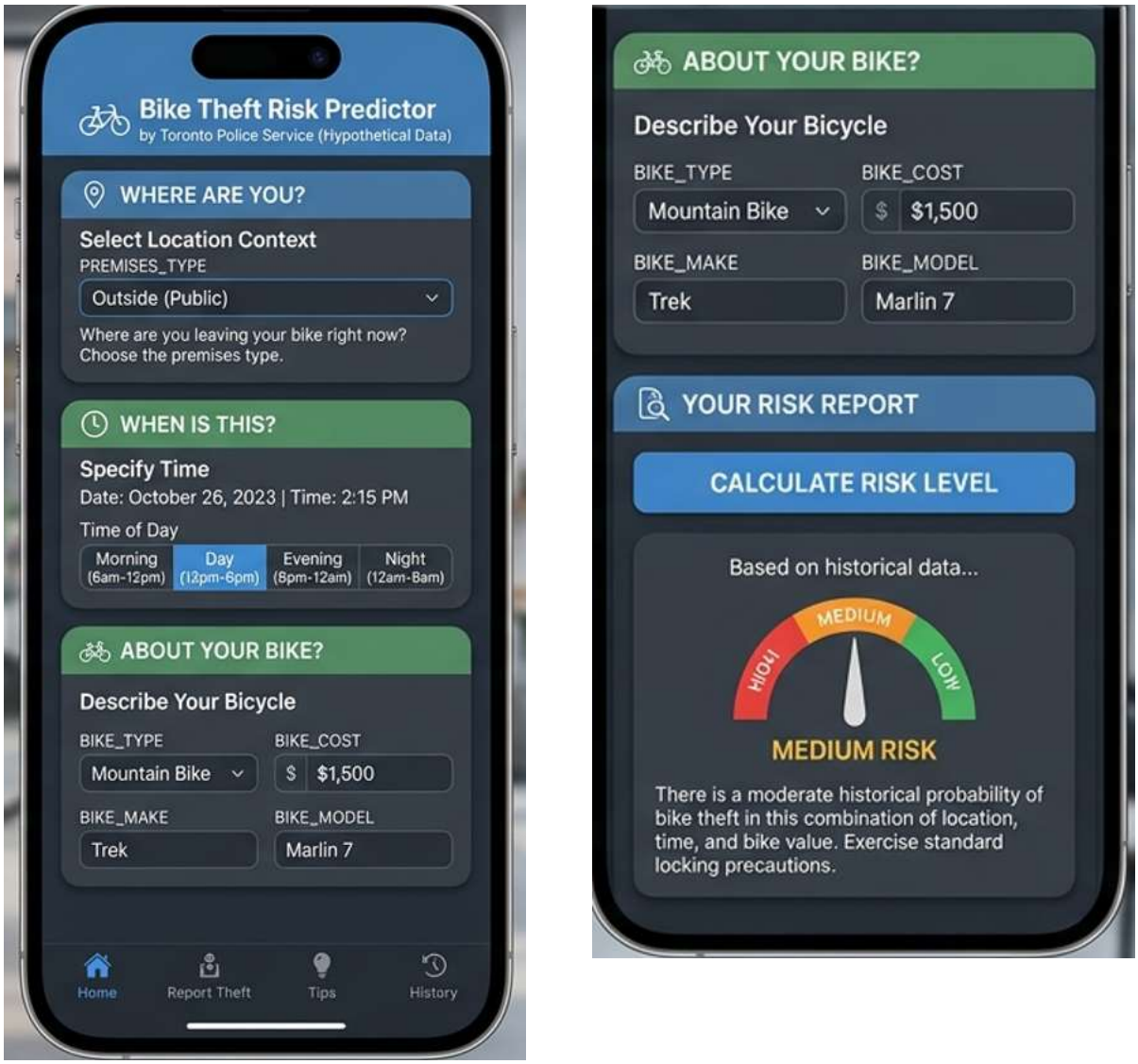


Figure 5- Mock-ups of "Bike Guard" app

Feature Engineering

To bridge the gap between raw data ingestion and model inference, three targeted feature engineering strategies were implemented, focusing on Spatial, Temporal, and Categorical dimensions of the Toronto Bike Theft dataset. These transformations convert high-variance data into structured, high-signal features, ensuring the Random Forest model can effectively identify patterns.

Neighborhood Risk Categorization

The transformation of raw theft counts into a categorical Risk Profile for each neighborhood allows for intuitive analysis of geographic safety trends and provides structured input for predictive modeling.

Methodology: Quantile-Based Segmentation

Neighborhoods are categorized using data-driven thresholds (33rd and 66th percentiles) for more relative classification:

- **High Risk**: Top 33% of neighborhoods by theft counts.
- **Medium Risk**: Middle 33%.
- **Low Risk**: Bottom 33%.
- **Unknown**: Areas with no recorded thefts.

Key Technical Insights

1. **Normalization**: Reduces the impact of extreme outliers, enhancing the analysis's robustness.
2. **Handling Sparsity**: Identifying neighborhoods with zero recorded thefts as “Unknown” ensures data integrity for future reports.
3. **Scalability**: Utilizing approxQuantile provides efficient scaling across large datasets while maintaining accuracy.

Business Value

This feature enables stakeholders to quickly identify high-priority zones, allowing for targeted patrols based on risk categorization.

Temporal Period Classification

Categorizing 24-hour incident timestamps into four distinct Occurrence Periods simplifies daily crime cycle analysis and helps allocate security resources effectively.

Methodology: Hour-Based Binning

The raw OCC_HOUR data is mapped into four key categories according to common daily activity patterns:

- Morning (06:00 - 12:00)
- Afternoon (12:01 - 17:00)
- Evening (17:01 -19:59)
- Night (20:00 - 05:59)

Key Technical Insights

1. **Logical Consistency**: Ensures all hours are accounted for, preventing data loss.
2. **Strategic Weighting**: Groups the lowest-light hours to identify peak theft windows.
3. **Efficiency**: High-level grouping facilitates easier visualization.

Business Value

This feature confirms peak theft times, enabling management to optimize security coverage according to hourly needs.

High-Cardinality Reduction (Bike Make/Model)

To manage the diverse range of bike manufacturers and models, high-cardinality distributions are simplified to focus on statistically significant entries.

Methodology: Top- N Filtering

Restricts entries to the 100 most frequent Makes and Models, categorizing any not on the list as “OTHER” to minimize noise.

Key Technical Insights

1. **Noise Reduction**: Concentrating on prevalent brands allows for focused analysis.
2. **Dimensionality Control**: Prevents unnecessary computational complexity during model training.
3. **Data Integrity**: Ensures clean schema for final analyses.

Business Value

This transformation allows for a clear "Market Share of Theft" analysis, highlighting key target brands without distractions.

Machine Learning

Two machine learning models were developed using a Random Forest classifier to predict bike theft risk levels.

Model A: With Location Features

This model was trained using all available location-based features.

Performance:

Accuracy: 100.00%

Precision: 100.00%

F1 Score: 100.00%

Confusion Matrix Summary (Model A)

The perfect classification indicates an over-reliance on spatial features, introducing potential data leakage (where 0-> High, 1-> Low, 2->Medium)

Label	Prediction	Count
0.0	0.0	410
1.0	1.0	353
2.0	2.0	394

Table 2- Information of the variable, prediction and counts for Model A

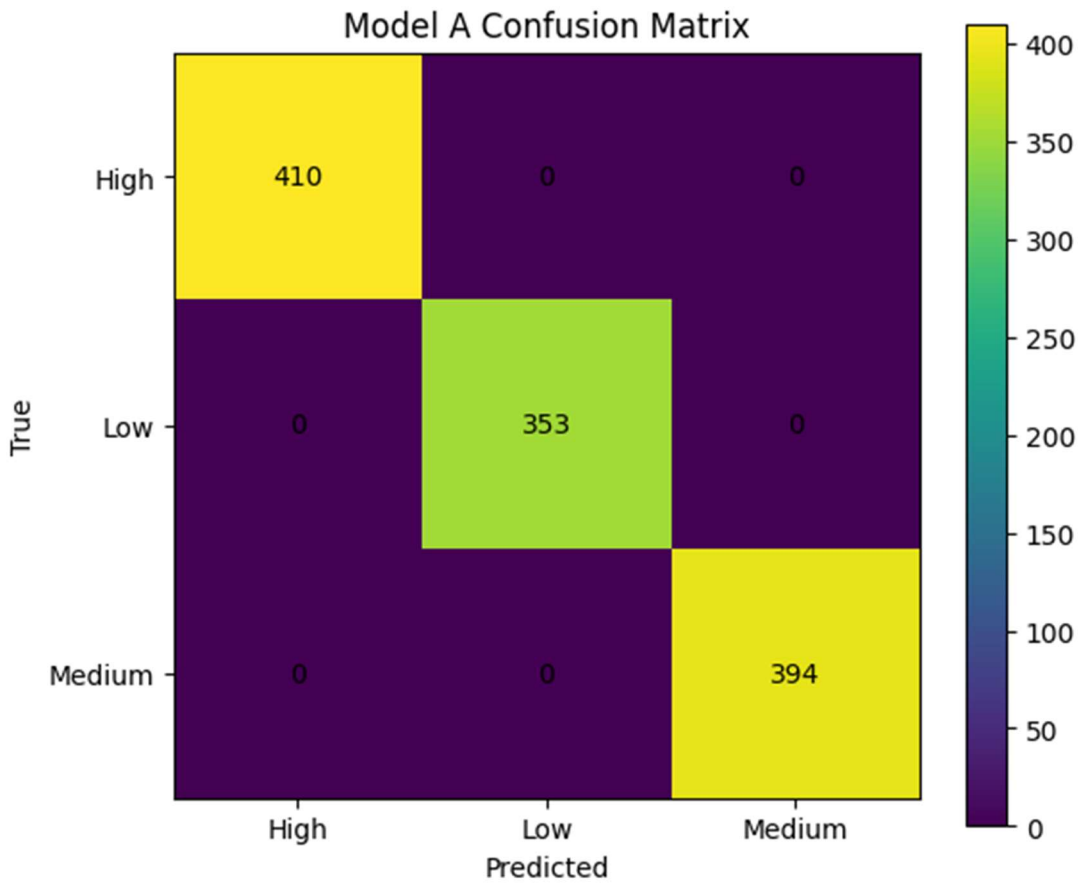


Figure 6- Heatmap of Confusion Matrix (Model A)

Model B: Without Location Features

By excluding location variables, Model B focuses on behavioral and temporal data.

Performance:

- Accuracy: 88.76%
- Precision: 90.48%
- F1 Score: 88.62%

Confusion Matrix Summary (Model B)

Model B provides a more realistic assessment of predictive performance. Without direct geographic indicators, the model must rely on underlying patterns in the data, such as temporal trends and other contextual features (where 0-> High, 1-> Low, 2->Medium).

Label	Prediction	Count
0.0	1.0	3
0.0	0.0	299
0.0	2.0	108
1.0	1.0	350
1.0	2.0	3
2.0	0.0	6
2.0	2.0	378
2.0	1.0	10

Table 3- Information of the variable, prediction and counts for Model B

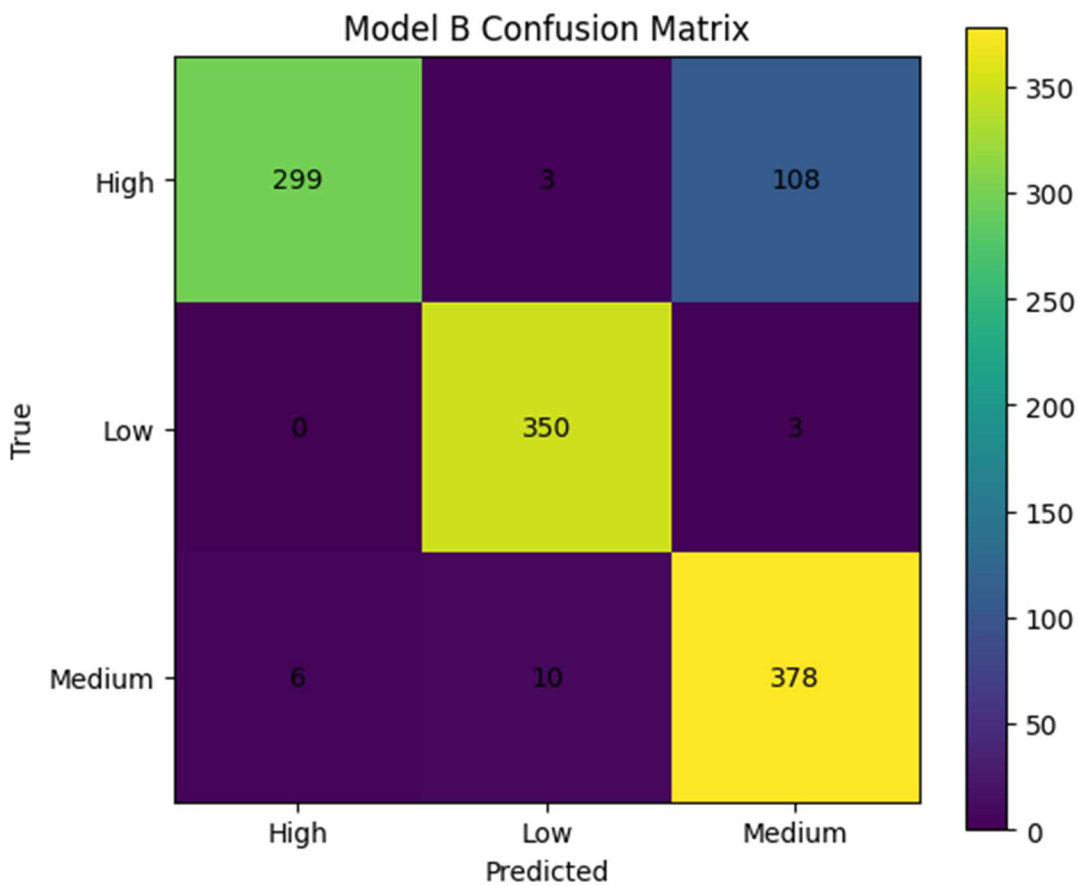


Figure 7- Heatmap of Confusion Matrix (Model B)

Model B Summary

While this model shows strong performance, it relies on underlying data patterns and avoids direct geographic indicators, making it suitable for real-world deployment.

Model Selection

Although Model A achieved perfect accuracy, it is unsuitable for deployment due to its data leakage. In contrast, Model B demonstrates robust performance and generalizability, making it the preferred choice for applications involving new data.

Simulating Real-Time Ecosystem: Clickstream Data Generation

To validate scalability and real-time processing capabilities, a synthetic data generator was created using PySpark, simulating thousands of concurrent users interacting with the application.

Key Simulation Characteristics

1. **Concurrent User Simulation**: 25,000 unique events simulate high-traffic scenarios.
2. **Contextual Realism**: Ensures statistical integrity by using actual values from the cleaned dataset.
3. **Temporal & Geospatial Mapping**: Events are categorized based on time-of-day logic and relevant police division data.
4. **Synthetic Reporting Lag**: Simulates realistic administrative delays for testing purposes.

Purpose in the Project

This stream serves as a stress test, measuring the risk-prediction model's performance under load and ensuring rapid risk assessments.

Technologies Used

The following technologies were employed in this project:

1. MongoDB (NoSQL)
2. SQLite (RDBMS)
3. Apache Spark
4. Spark Structured Streaming
5. Oracle VirtualBox
6. Ubuntu

Key Project Takeaways

1. Data wrangling is critical for extracting valuable insights and enhancing machine learning effectiveness.
2. Model comparison emphasizes the importance of careful feature selection and the risks of data leakage.
3. A comprehensive big data pipeline integrates cold processing, machine learning, and hot streaming.

Conclusion

This project successfully implements a big data pipeline leveraging the tools learned throughout the course and applying them to a production dataset. The analysis clearly shows that the highest volume of thefts occurs at night in apartment complexes along the waterfront. Consequently, the city of Toronto can enhance its policing strategies in these high-risk areas and utilize the “Bike Guard” app to help users assess risk by inputting vital bike details such as colour, make, model, and cost.

References

Sharma, D. (2026).
Week09_W26_Big_Data_System_Design_Lecture_Big_Data_Machine_Learning
[PowerPoint slides].https://learn.ontariotechu.ca/courses/37584/modules

Appendix

Screenshots of MongoDB queries and code is located here
https://github.com/AfeenaG/bike-theft-risk-prediction-lambda-architecture/tree/main/Code_Snippets